

DAOing It as a Collective: Designing the Future of Decentralised Personal Health Data Sharing

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ABSTRACT

The public depends on healthcare institutions to protect and govern medical-related information. However, with increasing security breaches and compliance failures with data protection law, researchers have begun to explore the feasibility of Decentralised Autonomous Organisations (DAOs) as a way of reimagining traditional forms of top-down data governance. In this paper, we describe early findings from the development of a card-based approach for engaging the public in the design of DAOs for collectively governing health data. Following a public workshop, we reflect on how our methodological approach involved laypeople in practical discussions about DAO design elements, such as voting mechanisms. Through group conversations, we observed how values affected the kinds of decentralised organisational structures participants wished to engage in. In particular, our analysis has implications for the future directions of DAO design, by pointing towards flexibility and modularity for voting and proposal interfaces, scalability and balancing power in tokenised communities.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods; HCI design and evaluation methods.**

KEYWORDS

Interaction Design, Decentralised Autonomous Organisations, Distributed Ledger Technology, Healthcare

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1 INTRODUCTION

With over 314 million patient records in the US exposed to data breaches between 2009 to 2021 [22], public trust in care providers'

abilities to protect personal health data is under threat. Distributed Ledger Technologies (DLTs) have been considered as an approach to mitigate growing data security and privacy challenges in healthcare. However, currently, traditional data governance models supervised by care institutions acting on behalf of patients, continue to dominate the healthcare sector [4]. Recent research [1][24][10] has demonstrated the technical viability of blockchain as a mechanism in which medical-related data could be controlled differently amongst patients, clinicians and care organisations, in particular to provide more privacy-minded structures [30]. Yet, in designing such decentralised systems, patient attitudes towards new forms of health data governance and their experiences of traditional healthcare services remain largely unknown and/or unaccounted for [17]. As a specific application of blockchain technology, decentralised autonomous organisations (DAOs) have recently emerged as a socio-technical framework, that enable remote online communities to operate as a cooperative organisation to achieve a particular shared purpose [15, 21]. DAOs typically are organisational or governance structures using DLT to automate codified rules, transaction processes and facilitate participation of members through voting mechanisms. With options for coordination and collective governance of resources, securely performed in accordance with pre-agreed rules defined in smart contracts[45], thus reviving debates around democratic participation in many areas[6, 35, 42, 44]. However, opportunities to understand, debate and shape the future design of DAOs in HCI have been restricted by a limited understanding [34] of appropriate public engagement strategies.

In this paper, we report on the development and evaluation of a card-based engagement methodology that aims to stimulate layperson debates on the design choices of DAOs as a mechanism to facilitate more cooperative forms of health data governance. Our two-phase development process comprised an initial review of pre-eminent DAOs and DAO builder platforms (e.g., DAOStack [14]) to establish a prior understanding of essential interactional elements that constitute a DAO. Informed by this understanding, we developed a set of feature cards and group tasks that were trialled through a half-day workshop involving seven members of the public. Specifically this late-breaking work contributes:

- (1) A methodological approach for lay publics to expand their understanding, and share viewpoints in the future design choices of DAOs, turning them into actionable user requirements for developers

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- (2) A reflection on an initial public engagement method and emerging design implications for a health data sharing DAO and its impact on wider HCI or Information Systems research

This study is an early exploration of an engagement approach that centres users and design aspects of DAOs for public debate. Although it was situated inside and hence somewhat limited to a healthcare data governance context, we believe this methodological design work is crucial to better understand public views regarding the deployment of data ownership and decentralised control models. These have the potential to transform a wide range of sectors, particularly since governments have begun to develop data trust legislation [32, 33]. In the following sections, we first provide background and root our contribution within a small but emerging field within human-computer interaction concerned with engaging publics in blockchain design. We also provide an overview of the evolving interaction components of DAOs, following an online review of DAO builder platforms. Secondly, we provide an overview of the development process undertaken in the creation of public engagement materials, including the definition of DAO feature cards, with a particular focus on proposal-making, incentivisation and voting actions typically performed through a DAO. Finally, we analyse and discuss early findings elicited through a public workshop.

2 DECENTRALISED TECHNOLOGIES, PUBLICS & DESIGN

Methods for encouraging public debates on emerging technologies such as DLTs have been developed in many contexts from nanotechnology in neuroscience [16] to biobanking in biomedical research [19]. Within the field of HCI, a small series of studies have looked to specifically engage publics in demystifying DLTs through novel methods and practices of design. For example, Maxwell et al. [31] developed a workshop toolkit named 'BlockExchange' that involved the use of 'resource' cards and Lego bricks as a material analogy of intangible 'blocks' within a decentralised public ledger to explore blockchain principles including immutability and value sharing via a trading game-like activity. Similarly, Khairuddin et al. [23] followed a material-centric approach to develop 'BlocKit', consisting of physical artefacts to support the designing and learning of blockchain infrastructure. Critically, such toolkits have aimed to support matters of infrastructural design, to materialise the intangible, uncover properties of blockchains and afford co-designers familiar abstractions in order to help them unpack complexities of distributed ledger network characteristics more generally.

Rankin et al.'s iterative exploration into the use of materiality and role-based gameplay for "designing tangible public engagements with DLTs and decentralised networks" [40] implemented a high-fidelity bodystorming engagement approach called 'PizzaBlock'. It looked to convey concepts of a blockchain network similar to BlockExchange and BlocKit, such as data flow and event transactions between stakeholders. It provided an opportunity to examine both infrastructural and specific interaction design concerns within a volunteering setting. Similarly, we have set out to open public engagement towards a specific blockchain-enabled technology (i.e., DAOs) within a well-defined application area (i.e., health data sharing) but crucially have, by design, sought to dissuade public discussion from the infrastructural aspects of blockchain systems,



Figure 1: DAO Design Elements

as previous research [34] into DLT design methods appear to have prioritised.

Nissen et al.'s research into "designing for distributed autonomous organisations" [38] represents a limited body of research that has explored the design implications of DAOs. Using materials to represent different types of relationships (e.g., wire, string, edible shoelaces) and entities (i.e., magnetic pins) within a DAO structure, the workshop encouraged groups of designers to ideate and physically map out decentralised organisations based on a chosen application area (e.g., a DAO for managing collectively owned housing) composed of relationships between "people, things and code" (ibid). In contrast, our work goes beyond attempts to ideate application possibilities of DAOs with designers and focuses on enabling public participants to experience and debate more pragmatic design issues such as the suitability of differing voting models in governing health data as a collective endeavour.

2.1 DAO Design Elements

To understand the foundational elements that constitute a DAO [35] prior to the development of our public engagement approach, we examined several emerging independent DAO builder platforms (i.e., Aragon, DAOstack and Colony) along with prominent DAO communities including 1Hive [48]. Given the lack of "compliance standards and established DAO use-cases" [2] at the time of our investigation, such platforms provided an initial basis on which to extrapolate common types of user interaction supported by DAOs. In particular, we gathered information from DAO builder documentation web portals and academic literature studies of independent DAO platforms to identify important areas of focus to assist in shaping our public design engagement outlined in Figure 1. These included core DAO concepts such as *proposal making*, *incentivisation* and *digital assets*.

A common user function universally offered across all DAO implementations we examined was *proposal voting* that enables voting on proposals made by organisation members. Specifically, we

Table 1: DAO Design Elements

Design Element	Description	Example Features
Proposal Making	Provides a process DAO members can utilise to develop ideas and suggest changes to the way an organisation operates in achieving its particular mission	Proposal anonymity, sponsorship, ranking
Digital & Physical Asset Management	Defines mechanisms for representing value in the form of digital tokens and its exchange within an online community	Token trading, vesting, staking & burning
Proposal Voting	Enables online individuals to express preference towards or veto community-led proposals	Super majority, quorum voting, delegated voting, cool-off periods, vetoing
Incentivisation	Represents incentive models in which members can redeem value for contributions made through cooperation within the community	Bounties, reputation rewards, voting power distribution
Arbitration & Dispute Resolution	Provides opportunities for members to resolve disputes and mediate conflicts	Proposal disputing, adjudication
Communication channels	Provides mechanisms for remote communications with members and to receive important updates	Social media (e.g., Discord), automated email notifications, reports

identified several voting related features, including various voting models (e.g., super majority, quorum & delegated voting), cooling off periods between voting decisions and proposals being actioned plus vetoing options. While this formative process was not intended to be comprehensive, it provided an opportunity to establish a coding structure for mapping DAO user functions to broader design concerns. In particular, this initial grounding was informed by previous systematic approaches that identified comparative feature sets of DAO builder platforms [2]. We identified six design elements of interest for debating DAOs with the public, along with several related features found in the implementations of current DAOs and builder platforms outlined in table 1. Critically, this mapping provided an initial basis to formulate our methodological approach.

3 METHODOLOGY

We set out to design a group-based engagement activity to elicit lay public debates on the design and use of DAOs for governing personal health data. As this research was conducted during a time of changing COVID-19 rules in the UK in early 2022 and our goal was to facilitate interaction in addition to discussions, we opted for an online-based workshop activity instead of a focus group. We took inspiration from previous studies as well as design thinking frameworks [5, 28, 46] to design tangible workshop materials beyond purely physical artifacts. Based on our formative grounding of DAO design elements and our previous research in the area of blockchain and health, including qualitative focus groups [37], we developed a series of materials including pre-activity information packs, a deck of feature cards and a series of workshop tasks. Before arranging a public workshop, we delivered the workshop activity to a small group of computing researchers with DLT expertise to evaluate our approach. This formative engagement highlighted shortcomings in the initial structure and timing of the planned workshop and related materials. For example, we simplified the description of particular feature cards for lay audiences.

The three-hour online workshop involved the recruitment of seven members of the public through the “Research for Future” Initiative supported by the National Institute for Health and Care

Research Clinical Research Network (NIHR-CRN). Due to the exploitative nature of our study, not aiming at data saturation or quantitative validity, the selection criteria was availability for the workshop date. Ahead of time, participants were sent materials to familiarise themselves with the topic. Our primary goal was to engage in debates about processes, values and preferences of lay persons, hence we did not focus on testing or querying participants’ understanding of DLT or DAOs. During the workshop, participants were divided into two groups with each given access to an online Miro board containing the cards and task sections that allowed participants to capture feature selections. Participants had the option to change, add or amend all their card configuration choices during the session. A member of the research team was present to assist with the Miro board and to address any questions of the participants. Our analysis is based on the conversations and video observations of the interactions with the Miro boards. Our results are based on qualitative and interpretive analysis of the group conversations and recorded videos of the interaction with the Miro boards [3, 12, 25].

3.1 Pre-activity Materials

The pre-activity material pack contained a narrative scenario (see appendix A) and the feature cards. The scenario was informed by our participation in research exploring public attitudes and user requirements on health data sharing with DLTs. During this process, data cooperatives have emerged as a possible governance model to support both individual and collective controls over health data donations. The scenario was the story of a group of fictitious persons with health problems who aim to set up a DAO to share and donate health data collectively with research, charitable or commercial entities. The basic idea of this potential DAO was to set up a platform for health data donations in which members can decide from the beginning how it is set up, individuals can contribute their personal data as well as make proposals on what to do with the data (e.g., who is allowed to access it), which will be put to a vote and decided. This data donation scenario put participants into a position in which they get to decide how this DAO is configured.

3.2 Feature Card Set

To make the workshop interactive and accessible for a lay audience with limited technical knowledge about blockchains or DAOs, we developed a card approach inspired by qualitative studies research into engaging publics in emerging technology debates [16, 27, 29, 36]. We created a set of cards, each consisting of a category, a title and an assigned colour (see examples in figure 2 and the full set in appendix B). There were five categories whose card content was derived from our formative examination of existing DAOs and DAO builder platforms, as discussed in the background section. In particular, our initial card set focused on voting, proposal creation, incentivisation models, values and different forms of member communication. There is a plethora of voting, proposal and incentive mechanisms within DAOs, often derived from liquid democracy [13, 26, 27]. We used a selection of common concepts (e.g., simple majority, super majority, a delegation of votes, bounties, etc.) as well as some esoteric ones (i.e., holographic consensus [14]). In order to design the cards more accessible to lay audiences, we simplified some concepts and added descriptions to most of them. For example, we “translated” the concept of holographic consensus into the “proposal ranking” (see Figure 2), as this is the essential function (ibid). Moreover, most DAOs use forums or on-line collaboration platforms like Mattermost, Slack and Discord to discuss everyday operations and upgrades of DAOs before they are implemented or voted on [41]. Therefore, we introduced communication as a category to explore the preferred ways of keeping in touch with the DAO and its members. The value cards were special, as they had no descriptions and consisted of values propositions promoted within the blockchain and DAO communities, such as equality, fairness, trust and security [17]. While most DAOs have mission statements promoting certain values, these cards’ primary function was argumentation aids for participants [16] and to give us a richer understanding of their configuration choices and card selections in relation to the workshop tasks.

3.3 Workshop Tasks

The main workshop activity was focused on tasks that had the dual purpose of adhering to the framework of the data donation scenario while also motivating participants to discuss DAO design elements in an engaging manner. The cards were divided into four tasks, in which participants would choose at least one card per category to configure their imaginary DAO. The selection of tasks emerged through previous research on collective health data sharing paired with issues of DAO participation identified in section 2.1. The first task was based on the question “How should the collective make and vote on proposals and why?” using the voting and proposal-making cards. The second task was to focus on incentive cards as a means of dealing with low participation and voter turnout, a common phenomenon within DAOs [11, 20, 43]. The third task was around values and the question of how power should be distributed within tokenised or data-driven communities [47]. The last task was to determine the communication channels participants would prefer to interact with the DAO and its members.

4 FINDINGS & DISCUSSION

Our experiences of developing a method for engaging publics in design debates regarding DAOs and observation of public interactions during a half-day workshop has provided an early understanding of how the materials influenced group discussions. We were interested in understanding two aspects: (1) Methodological considerations of our engagement approach for DAO design. (2) How the approach worked in relation to health data governance and its design implications.

To understand the broader methodological considerations for DAO design approaches, we first were interested in how the method and materials influence group discussions and behaviours. Overall, the method worked to stimulate discussions of complex DAO arrangements without prior knowledge of the distributed ledger technologies and to iterate on a pre-given set of configuration options to capture user requirements. The Miro boards aided the level of interactivity amongst the groups even though the session was held online, providing a collaborative workspace where the tasks, the cards and representation of the configuration choices were visible and accessible to each participant. The ability to drag the cards in and out of the middle circle (see figure 2) aided the discussion process by providing a tangible goal to work towards. Being able to use sticky notes to add missing configuration choices and/or amend existing cards enabled the participants to extend the initial set of cards provided.

Further, we questioned how our approach affect laypeople’s participation in and conversations regarding DAO design features in relation to the governance of health data. Although participants were given access to the cards before the workshop, some had trouble reading the card descriptions on their screen, with one participant opening the cards on a separated tablet device. This resulted in participants collectively reading the cards and tasks out loud. Although most of the participants found the cards and tasks easy to grasp, some insisted on more contextual information. For example, Group A directed questions toward the researcher, resulting in a constant back and forth, turning the intended group activity into a group interview. In contrast, group B appeared to discuss the tasks and cards among themselves, with minimal clarification from the researcher. Despite these differences, each group engaged in similar design discussions and chose many of the same cards. This suggests that the cards and tasks provided enough framing to support different communication styles while avoiding significant deviation as a result of the additional context provided by the researchers [18]. We found that DAO design element discussions that were intended to be separated by each of the four tasks specifically debate on proposal making, voting interactions and incentivisation models, occurred during the initial task debate on aspects of proposal making. That involved the use of cards highlighting proposal anonymity, ranking & sponsorship features. Moreover, we observed participants’ use of the cards as a tool for instigating moral discussions regarding the organisational structure of a DAO for health data sharing rather than specific design discussions around the features presented. The question of whether it is form over function or if form is function echos findings of the BlocKit study, which stated: “[...]the mental models that people develop for [complex technologies] are likely to differ based on their specific forms of interaction and goals.” [23].



Figure 2: Example snapshot of group A's Miro board showing proposal and voting cards as well as annotations made by participants. Participants were encouraged to drag cards into the blue circle are for selection for the round-based tasks. High resolution image of cards and Miro board can be found in supplement materials.

Finally, we reflected on how the materials and tasks be improved upon. Our early findings indicate that the cards and materials provided an effective mechanism for encouraging participants to form points of view toward the use of DAOs in the governance of personal health data more broadly. However, participants appeared less drawn toward more pragmatic debates that reflected on the practical implications of differing design element choices outlined on the cards. This would suggest a need to further refine our approach to guide group discourse towards more practical design thinking in order to better understand group reasoning around the final card choices that were made. However, in general, our findings demonstrate that participants could have productive debates and begin to formulate opinions on DAOs in a healthcare context.

4.1 DAO Design Implications

Previous design work engaged on the general or infrastructural aspects of blockchain [23, 31, 34], in contrast, our approach focused on opening up layperson discussions of DAO design features in a specific domain of healthcare. Indeed, we observed minimal debates around the technical aspects of the underlying distributed ledger technology [40] of DAOs. Through this first iteration of a user-centred methodological approach to support the engagement of publics with DAOs [38] in health data management, we aimed to analyse prominent DAO implementations and map out core features to offer a representative view of the design landscape, placing a particular focus on technical accuracy when explaining

and engaging in DAO design with healthcare users. The pre-activity information material and framing of the data donation scenario allowed for in-depth discussions about health data governance as it illustrated a collective point of reference participants could use when exploring card options during each task. As a direct consequence, the cards were always debated within the given scenario, limiting the card choices to health governance. Nevertheless, the cards appeared to provide participants with enough information on possible configuration options to produce a draft of a DAO in each group. The focus on individual and collective health data governance set the groups up to deal with normative decisions and trade-offs between autonomy of individual members, power (im)balances and enabling possibilities for participation. Decisions for or against configuration options were always framed within these value trade-offs and negotiations as to what participants saw as acceptable forms of intra-DAO hierarchies. Our analysis showed that while the two groups made similar choices overall for their DAO configuration, there were some differences in their final card choices and the groups' internal deliberation process. Variation in group choices rather reflected on preferences and processes within the DAO. This highlights the need to design for specific group needs. In sum, we observed the following patterns and design implications:

4.1.1 Modularity & Adaptability of DAOs. Participants agreed that organisational structures must be adaptable and changeable in response to a variety of factors. Participants perceived larger organisations as less efficient, so scalability of decision-making emerged as a major issue. Proposals were made to limit the number of members in the DAO, revoke membership if members were inactive for a set period of time, and then redistribute voting power among the remaining members. Delegation options (e.g., an elected trustee) were also mentioned for those who do not or cannot actively participate or have no knowledgeable expertise. The multiple choices of cards in both groups, as well as their additions of notes, demonstrated how they must satisfy two needs: (1) fluidity to adapt to specific settings and proposals, but also (2) providing an a priori structure that reflects values on fairness and equality.

4.1.2 Templates, Formulation Aids, or Standards for Proposal Creation. Both groups preferred proposal ranking as a method of filtering and prioritising proposals, thereby assisting members in managing proposals. Several participants used their personal and professional experiences to argue that using specific language and rich descriptions was critical to the success of a proposal being passed. To assist members in expressing their ideas, they desired aids or standards for creating proposals in the first place. Others debated that the voting mechanism is quite situational and that there may need to be community rules governing which voting mechanism will be used for what type of proposal, resulting in the creation of templates that members can use.

4.1.3 Mechanisms for Voting, Incentivising & Limiting Individual Member's Power. During the group discussions, two issues arose that dominated the final decisions and amendments: First, there is the possibility of fraudulent activities, particularly malicious entities flooding the DAO with fake accounts in order to influence voting decisions solely in their favour (e.g., companies who might

aim to access data). The second issue of concern was the concentration of power among individual members. Concerns were raised that a powerful individual could syphon off funds and fail to uphold particular values reflected through the DAO. Participants proposed limiting the power of individuals above a certain accumulation and creating tier systems for new members on voting power accumulation, such as time lengths or participation locks. Similar debates raged in groups about financial rewards as incentives for individuals contributing data, with some questioning whether it would reduce members' involvement to a purely transactional nature for monetary gain. Participants expressed preference towards incentive models that would allow members to develop and demonstrate online social capital through reputation scores. However, debates centred on the importance of voting and incentive mechanisms that promoted solidarity amongst collective members and embraced altruistic values.

5 CONCLUSION

Our investigation has led to the development of an engagement activity that aimed to better understand public attitudes towards emerging forms of decentralised data governance, and involve care users in open debates that encouraged them to consider core features and accordances of DAOs as enabling technology in the collective management of personal health data. Moreover, we have examined DAO builder frameworks to provide an early grounding of the DAO design space, highlighting six interaction design elements (e.g., proposal making, voting & incentives) that informed the primary contribution of this late-breaking research, the development of a public workshop activity comprising use-case scenarios, feature cards and design tasks. Following an online public workshop, we provide healthcare practitioners and interaction designers early insights into participant group practices in the deliberation of DAO design choices, through a card-based approach for conveying complex DAO concepts to lay audiences. In addition, we contextualised reflections on the use of our method with suggestions from participants that emerged as part of the design discussions, such as desires for adaptive voting features for different forms of proposal making. Finally, this formative work highlights ways in which we adapted our approach to mitigate the challenges of the COVID-19 pandemic and provides a foundation for future work on public engagements with DAOs. Further research could include using the card-based approach on professional users or groups starting DAOs. Beyond DLT applications, this research approach could also be used within other contexts, particular with a focus on decentralisation [7], data trusts, and those that require negotiations on design aspects such as modularity [8, 9], personalised medicine, or governance and provision of information [39].

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REFERENCES

- [1] Cornelius C. Agbo, Qusay H. Mahmoud, and J. Mikael Eklund. 2019. Blockchain technology in healthcare: a systematic review. *Healthcare* 7 (2019), 30. Issue 56.
- [2] E. Banimemeh, S. Farshidi, and S. Jansen. 2023. A decision model for decentralized autonomous organization platform selection: Three industry case studies. *Blockchain: Research and Applications* 100127 (2023), 24. <https://doi.org/10.1016/j.bcr.2023.100127>
- [3] Rosaline Barbour. 2008. *Doing Focus Groups*. SAGE, London, UK. Google-Books-ID: TzZTCEAK6N4C.
- [4] Mahmood A. Bazel, Fathey Mohammed, and Mazida Ahmed. 2021. Blockchain Technology in Healthcare Big Data Management: Benefits, Applications and Challenges. In *2021 1st International Conference on Emerging Smart Technologies and Applications (eSmarTA)*. IEEE, Sana'a, Yemen, 1–8. <https://doi.org/10.1109/eSmarTA52612.2021.9515747>
- [5] Erling Björgvinsson, Pelle Ehn, and Per-Anders Hillgren. 2012. Design Things and Design Thinking: Contemporary Participatory Design Challenges. *Design Issues* 28, 3 (2012), 101–116. https://doi.org/10.1162/DESI_a_00165
- [6] Jaya Klara Brekke. 2020. Hacker-engineers and Their Economies: The Political Economy of Decentralised Networks and 'Cryptoeconomics'. *New Political Economy* 0, 0 (Aug. 2020), 1–14. <https://doi.org/10.1080/13563467.2020.1806223> Publisher: Routledge _eprint: <https://doi.org/10.1080/13563467.2020.1806223>
- [7] Jean-Paul Calbimonte, Orfeas Aidonopoulos, Fabien Dubosson, Benjamin Pocklington, Ilia Kebets, Pierre-Mikael Legris, and Michael Schumacher. 2023. Decentralized semantic provision of personal health streams. *Journal of Web Semantics* 76 (April 2023), 100774. <https://doi.org/10.1016/j.websem.2023.100774>
- [8] Francisco Maria Calisto, Nuno Nunes, and Jacinto C. Nascimento. 2020. BreastScreening: On the Use of Multi-Modality in Medical Imaging Diagnosis. In *Proceedings of the International Conference on Advanced Visual Interfaces (AVI '20)*. Association for Computing Machinery, New York, NY, USA, 1–5. <https://doi.org/10.1145/3399715.3399744>
- [9] Francisco Maria Calisto, Carlos Santiago, Nuno Nunes, and Jacinto C. Nascimento. 2021. Introduction of human-centric AI assistant to aid radiologists for multimodal breast image classification. *International Journal of Human-Computer Studies* 150 (June 2021), 102607. <https://doi.org/10.1016/j.ijhcs.2021.102607>
- [10] Janya Chanchaichujit, Albert Tan, Fanwen Meng, and Sarayoot Eaimkhong. 2019. Blockchain Technology in Healthcare. In *Healthcare 4.0: Next Generation Processes with the Latest Technologies*, Janya Chanchaichujit, Albert Tan, Fanwen Meng, and Sarayoot Eaimkhong (Eds.). Springer Singapore, Singapore, 37–62. https://doi.org/10.1007/978-981-13-8114-0_3 00000.
- [11] Mengting Chang, Qingfei Min, and Zipei Li. 2019. Understanding Members' Active Participation in a DAO: An Empirical Study on Steemit. In *PACIS 2019 Proceedings*, Vol. 197. AIS Electronic Library (AISeL), X'ian, China, 9.
- [12] Christian Heath, Jon Hindmarsh, and Paul Luff. 2010. *Analysing Interaction: video, ethnography and situated conduct*. SAGE Publications, Inc., London. <https://www.semanticscholar.org/paper/Analysing-Interaction%3A-video%2C-ethnography-and-Heath-Hindmarsh/cbc8635a6fc54597c680e7d17439ab63ba35a70c>
- [13] Amrita Dhillon, Grammateia Kotsialou, Peter McBurney, and Luke Riley. 2019. *Introduction to Voting and the Blockchain: some open questions for economists*. CAGE Online Working Paper Series. Competitive Advantage in the Global Economy (CAGE). <https://econpapers.repec.org/paper/cgewacage/416.htm>
- [14] Youssef Faqir-Rhazoui, Javier Arroyo Gallardo, Samer Hassan, Youssef Faqir-Rhazoui, Javier Arroyo Gallardo, and Samer Hassan. 2021. A Scalable Voting System: Validation of Holographic Consensus in DAOstack. <https://eprints.uct.ac.za/eprint/62303/>
- [15] Manfred Faßler. 2006. Communities of Projects. In *Lesebuch Projekte: Vorgriffe, Ausbrüche in die Ferne*, Christian Reder (Ed.). Springer, Vienna, 141–169. https://doi.org/10.1007/3-211-38096-5_9
- [16] Ulrike Felt, Simone Schumann, Claudia G. Schwarz, and Michael Strassnig. 2014. Technology of imagination: a card-based public engagement method for debating emerging technologies. *Qualitative Research* 14, 2 (2014), 233–251. <https://doi.org/10.1177/1468794112468468> _eprint: <https://doi.org/10.1177/1468794112468468>
- [17] Marcus Foth. 2017. The promise of blockchain technology for interaction design. In *Proceedings of the 29th Australian Conference on Computer-Human Interaction (OZCHI '17)*. Association for Computing Machinery, New York, NY, USA, 513–517. <https://doi.org/10.1145/3152771.3156168>
- [18] Paul Galdas. 2017. Revisiting Bias in Qualitative Research: Reflections on Its Relationship With Funding and Impact. *International Journal of Qualitative Methods* 16, 1 (2017), 1609406917748992. <https://doi.org/10.1177/1609406917748992> Publisher: SAGE Publications Inc.
- [19] Melanie Goisauf and Anna P. Durnová. 2019. From engaging publics to engaging knowledges: Enacting "appropriateness" in the Austrian biobank infrastructure. *Public Understanding of Science* 28, 3 (April 2019), 275–289. <https://doi.org/10.1177/0963662518806451> Publisher: SAGE Publications Ltd.
- [20] Eduardo Gómez. 2016. The DAO Undergoes Low Voting Turnout. <https://themerkle.com/the-dao-undergoes-low-voting-turnout/>
- [21] Samer Hassan and Primavera De Filippi. 2021. Decentralized Autonomous Organization. *Internet Policy Review* 10, 2 (2021), 1–10. <https://doi.org/10.14763/>
- [22] HIPAA. 2022. Healthcare Data Breach Statistics - Latest Data for 2022. <https://www.hipaajournal.com/healthcare-data-breach-statistics>
- [23] I.E. Khairuddin, C. Sas, and C. Speed. 2019. BlockKit: A physical kit for materializing and designing for blockchain infrastructure. In *Proceedings of the 2019 ACM Designing Interactive Systems Conference*. ACM, San Diego, CA, USA, 1449–1462. <https://doi.org/10.1145/3322276.3322370>
- [24] Seyednima Khezr, Md Moniruzzaman, Abdulsalam Yassine, and Rachid Benlamri. 2019. Blockchain technology in healthcare: A comprehensive review and directions for future research. *Applied Sciences (Switzerland)* 9, 9 (2019), 28. <https://doi.org/10.3390/app9091736>
- [25] Ulrike Kissmann (Ed.). 2009. *Video interaction analysis: methods and methodology*. Peter Lang, Frankfurt am Main ; New York. OCLC: 311755462.
- [26] Grammateia Kotsialou and Luke Riley. 2020. Incentivising Participation in Liquid Democracy with Breadth-First Delegation. In *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (Auckland, New Zealand) (AAMAS '20)*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 638–644.
- [27] Yong Ming Kow and Caitlin Lustig. 2018. Imaginaries and Crystallization Processes in Bitcoin Infrastructure. *Computer Supported Cooperative Work (CSCW)* 27, 2 (April 2018), 209–232. <https://doi.org/10.1007/s10606-017-9300-2>
- [28] Thomas Lockwood. 2009. *Design Thinking: Integrating Innovation, Customer Experience, and Brand Value*. Allworth Press, New York, USA. <https://openlibrary.telkomuniversity.ac.id/pustaka/117516/design-thinking-integrating-innovation-customer-experience-and-brand-value.html>
- [29] Caitlin Lustig. 2019. Intersecting Imaginaries: Visions of Decentralized Autonomous Systems. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (Nov. 2019), 210:1–210:27. <https://doi.org/10.1145/3359312>
- [30] Andrea Margheri, Massimiliano Masi, Abdallah Miladi, Vladimiro Sassone, and Jason Rosenzweig. 2020. Decentralised provenance for healthcare data. *International Journal of Medical Informatics* 141 (Sept. 2020), 104197. <https://doi.org/10.1016/j.ijmedinf.2020.104197>
- [31] Deborah Maxwell, Chris Speed, and Dug Campbell. 2015. 'Effing' the ineffable: opening up understandings of the blockchain. In *Proceedings of the 2015 British HCI Conference (British HCI '15)*. Association for Computing Machinery, New York, NY, USA, 208–209. <https://doi.org/10.1145/2783446.2783593>
- [32] McFarlane Ben. 2019. Data Trusts and Defining Property. <https://www.law.ox.ac.uk/research-and-subject-groups/property-law/blog/2019/10/data-trusts-and-defining-property>
- [33] Richard Milne, Annie Sorbie, and Mary Dixon-Woods. 2021. What can data trusts for health research learn from participatory governance in biobanks? *Journal of Medical Ethics* 48, 5 (2021), 6. Publisher: Institute of Medical Ethics.
- [34] Dave Murray-Rust, Chris Elsdén, Bettina Nissen, Ella Tallyn, Larissa Pschetz, and Chris Speed. 2021. Blockchain and Beyond: Understanding Blockchains through Prototypes and Public Engagement. *ACM Transactions on Computer-Human Interaction* 29, 5 (Dec. 2021), 73. <https://doi.org/10.1145/3503462> arXiv: 2112.11891.
- [35] Kelsie Nabben. 2022. DAO Design Patterns: Components that constitute "Decentralized Autonomous Organizations". <https://medium.com/block-science/dao-design-patterns-1193a694fb55>
- [36] Lisa P. Nathan, Predrag V. Klasnja, and Batya Friedman. 2007. Value scenarios: a technique for envisioning systemic effects of new technologies. In *CHI '07 Extended Abstracts on Human Factors in Computing Systems (CHI EA '07)*. Association for Computing Machinery, New York, NY, USA, 2585–2590. <https://doi.org/10.1145/1240866.1241046>
- [37] Victoria Neumann, Gail Davidge, Mike Harding, James Cunningham, Nigel Davies, Sarah Devaney, Gary Leeming, Søren Holm, and John Ainsworth. 2023. Examining public views on decentralised health data sharing. *PLOS ONE* 18, 3 (March 2023), e0282257. <https://doi.org/10.1371/journal.pone.0282257> Publisher: Public Library of Science.
- [38] Bettina Nissen, Ella Tallyn, and Kate Symons. 2019. Tangibly understanding intangible complexities: Designing for distributed autonomous organizations. *Ubiquity: The Journal of Pervasive Media* 6, 1 (Nov. 2019), 47–63. https://doi.org/10.1386/ubiq_00007_1
- [39] Mikael Ohrling, Karin Solberg Carlsson, and Mats Brommels. 2022. No man is an island: management of the emergency response to the SARS-CoV-2 (COVID-19) outbreak in a large public decentralised service delivery organisation. *BMC Health Services Research* 22, 1 (March 2022), 371. <https://doi.org/10.1186/s12913-022-07716-w>
- [40] Jonathan Rankin, Chris Elsdén, Ian Sibbald, Alan Stevenson, John Vines, and Chris Speed. 2020. PizzaBlock: Designing artefacts and roleplay to understand decentralised identity management systems. In *DIS '20: Proceedings of the 2020 ACM Designing Interactive Systems Conference*. ACM Designing Interactive Systems 2020, Eindhoven, Netherlands, 1593–1606. <https://doi.org/10.1145/3357236.3395568>
- [41] Ellie Rennie, Michael Zargham, Joshua Tan, Luke Miller, Jonathan Abbott, Kelsie Nabben, and Primavera De Filippi. 2022. Toward a Participatory Digital Ethnography of Blockchain Governance. *Qualitative Inquiry* 28, 7 (2022),

10778004221097056. <https://doi.org/10.1177/10778004221097056>
- [42] Nathan Schneider. 2019. Decentralization: an incomplete ambition. *Journal of Cultural Economy* 12, 4 (July 2019), 265–285. <https://doi.org/10.1080/17530350.2019.1589553> Publisher: Routledge _eprint: <https://doi.org/10.1080/17530350.2019.1589553>.
- [43] The Ontology Team. 2021. Using Reputation in DAO Governance. <https://medium.com/ontologynetwork/using-reputation-in-dao-governance-7307eb3f3827>
- [44] Christopher Tozzi. 2019. Decentralizing democracy: approaches to consensus within blockchain communities. *Teknokultura. Revista de Cultura Digital y Movimientos Sociales* 16, 2 (Oct. 2019), 181–195. <https://doi.org/10.5209/tekn.64523>
- [45] Shuai Wang, Liwei Ouyang, Yong Yuan, Xiaochun Ni, Xuan Han, and Fei-Yue Wang. 2019. Blockchain-Enabled Smart Contracts: Architecture, Applications, and Future Trends. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 49, 11 (Nov. 2019), 2266–2277. <https://doi.org/10.1109/TSMC.2019.2895123> Conference Name: IEEE Transactions on Systems, Man, and Cybernetics: Systems.
- [46] Mikael Wiberg. 2014. Methodology for materiality: interaction design research through a material lens. *Personal and Ubiquitous Computing* 18, 3 (2014), 625–636. <https://doi.org/10.1007/s00779-013-0686-7>
- [47] Yongjing Xie, Zhihong Li, and Xiaoying Xu. 2021. Addressing Wealth Inequality Problem in Blockchain-Enabled Knowledge Community with Reputation-Based Incentive Mechanism. In *ICEB 2021 Proceedings*. AIS Electronic Library (AISeL) AIS Electronic Library (AISeL), Nanjing, China, 12.
- [48] Michael Zargham and Kelsie Nabben. 2022. Aligning 'Decentralized Autonomous Organization' to Precedents in Cybernetics. <https://doi.org/10.2139/ssrn.4077358>